11th tutorial

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1. Circular Shift Let A be an array of n distinct integers which are sorted into increasing order and then circularly shifted by k positions to the right. For instance, $[3, 4, 8, 11, 15, 18]$ is sorted and shifted by $k = 3$ into $[11, 15, 18, 3, 4, 8]$.

Given such an array which has been shifted by an unknown k, describe a logarithmic-time algorithm that determines the value of $k \bmod n$.

2. Inversions Given a permutation A of $\{1, 2, \ldots, n\}$ and two indices $i < j$ we say that i and j are inverted if $a_i > a_j$. For example, $A = [5, 3, 4, 1, 2]$ has eight inverted pairs: $(1, 2)$, $(1, 3)$, $(1, 4)$, $(1, 5)$, $(2, 4)$, $(2, 5)$, $(3, 4)$ and $(3, 5)$.

Describe an algorithm which counts the number of inverted index-pairs in a given permutation with time complexity $\mathcal{O}(n \log n)$.

3. Twisted cable We have a long data cable with n individual wires exposed on either end; each wire on the left end is connected to exactly one wire on the right end. We need to find out which wires map to which in order to transmit data efficiently, but the cable has been twisted, so this mapping is not obvious.

We can use the following operations:

(a) connect electricity to a given wire on the left end,

(b) disconnect electricity from a given wire on the left end and

(c) measure electricity on a given wire on the right end.

Design an algorithm to finds out which wire is connected to which while minimizing the number of these operations.

4. Nuts And Screws At the input, we are given n nuts and n bolts of different sizes. For each bolt, there is at least one suitable nut but we can only compare one nut to one bolt at a time. Doing so tells us that either: the nut is too small, the nut is too large or the nut and bolt fit together.

Describe an algorithm which pairs nuts and bolts as quickly as possible. A randomized solution with average time $O(n \log n)$ is enough.